Indian Institute of Information Technology Tiruchirappalli

3rd Year (CS515 - Unix Programming) - Assignment 03

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Total Marks: 100

Deadline: 29-03-2026

Objective:

To demonstrate the practical proficiency in the usage of version control (Git), containerization (Docker), and container orchestration (Kubernetes) by developing and deploying a multi-tier application.

Problem Statement:

For this assignment, you will build a basic web application of your choosing. The only requirement for the app itself is that it must have two connected parts: a web interface (what the user sees) and a database (to store information). Your main task is not just writing the code, but using **Git** to track your progress, **Docker** to package your app, and **Kubernetes** to run it on your local machine.

The assignment is divided into three phases:

Phase 1: Application Development & Version Control (Git)

○ Develop a two-tier application consisting of a web front-end/API and a backend service (example: a database like Redis, PostgreSQL, or MongoDB).

○ Initialize a Git repository for your project.

○ No Monolithic Commits: You must commit incrementally. For example, separate commits for “Initial Project Structure”, “Added database connection logic”, and “Updated README”. A single “Finished project” commit is unacceptable. Your commit history must show progressive development.

Phase 2: Containerization (Docker)

○ Write a **Dockerfile** for your application.

○ The image should be optimized (example: using a multi-stage builds if applicable) and pull successfully.

○ Create a **docker-compose.yml** file to run your application and its dependencies locally.

Phase 3: Orchestration (Kubernetes)

○ Environment: Use **Minikube** as your Kubernetes cluster.

○ Write Kubernetes manifest files to deploy your application to a local cluster.

○ Required Manifests:

- Deployment (for your application, with at least 2 replicas).
- Service (to expose your application, example: NodePort or LoadBalancer).

Submission Requirements:

1. You must submit a single PDF document that acts as the complete guild and report for your project.

2. Your PDF must include:

○ The GitHub Link: A working link to your public GitHub repository. (Your repository must contain all of your source code, your **Dockerfile**, and your **Kubernetes .yaml** files).

○ Project Overview: A brief, simple explanation of what your application does.

○ Step-by-Step Instructions: The exact terminal commands and steps required to build your Docker image and apply your Kubernetes files to get the application running.

○ Proof of Execution (Screenshots): You must paste the following screenshots in your PDF:

- The output of ***`git log –oneline`*** in your terminal (to show your commit history).
- The output of ***`kubectl get pods`*** (to prove your containers are actually running).
- A screenshot of your final application successfully running in your web browser.